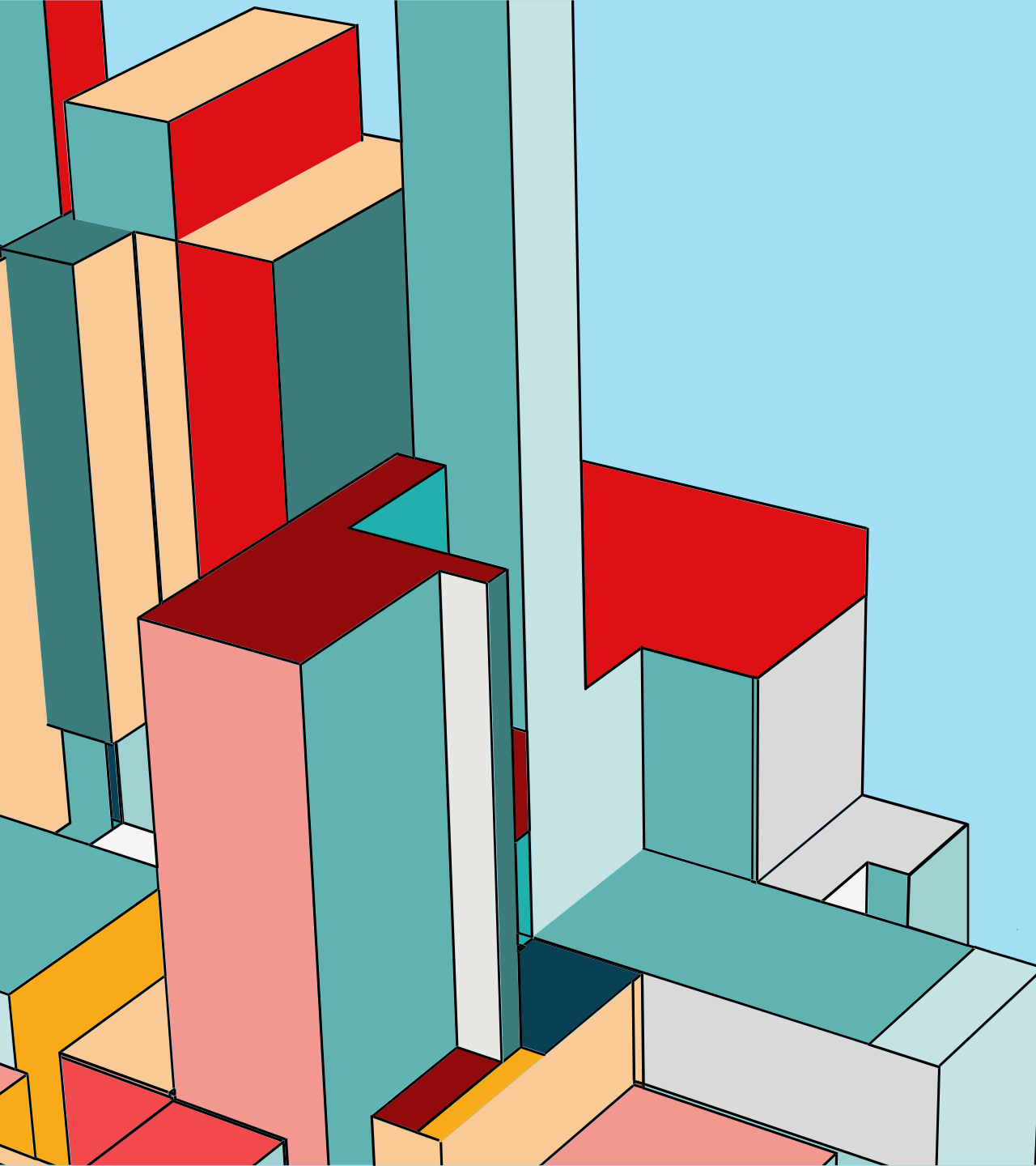


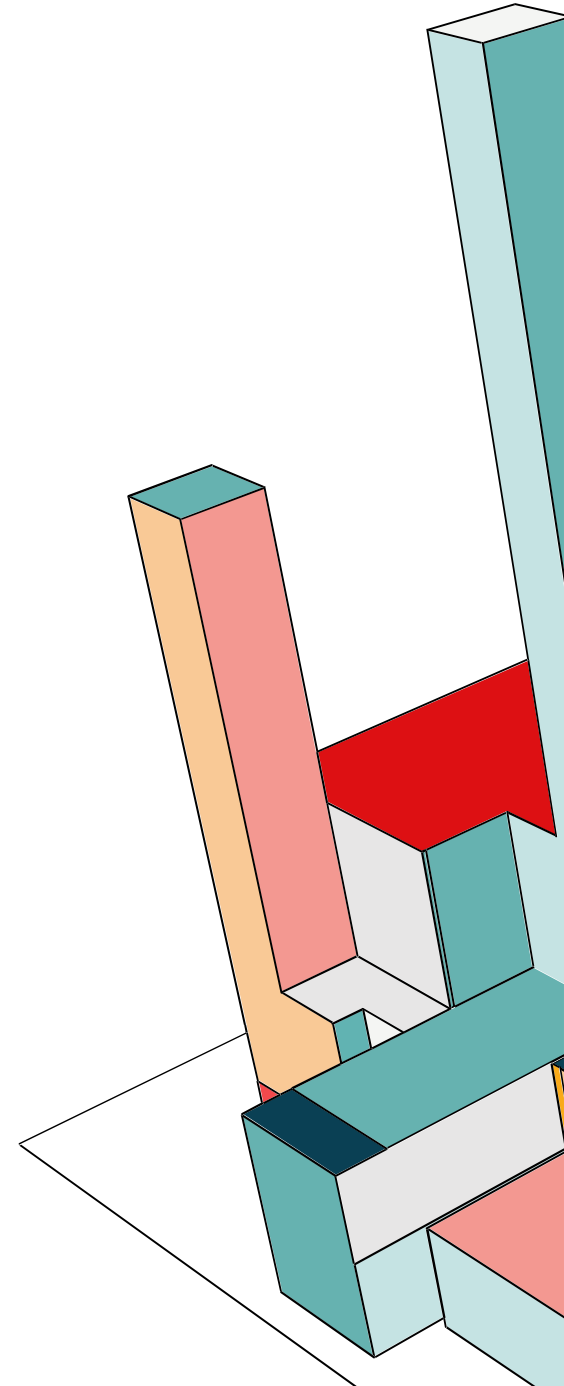


“COMPANY” DATA ANALYSIS REPORT

“jan - feb - Mar”

METHOD APPROACH

1. Data Manipulation and Cleaning
2. Analysis & Visualization
3. Dashboard
4. Automation
5. Rate Simulation
6. Answering Business Questions
7. Insights & Recommendation





DATA MANIPULATION & CLEANING

DATA MANIPULATION & CLEANING

1. Cleaning raw data Column renaming

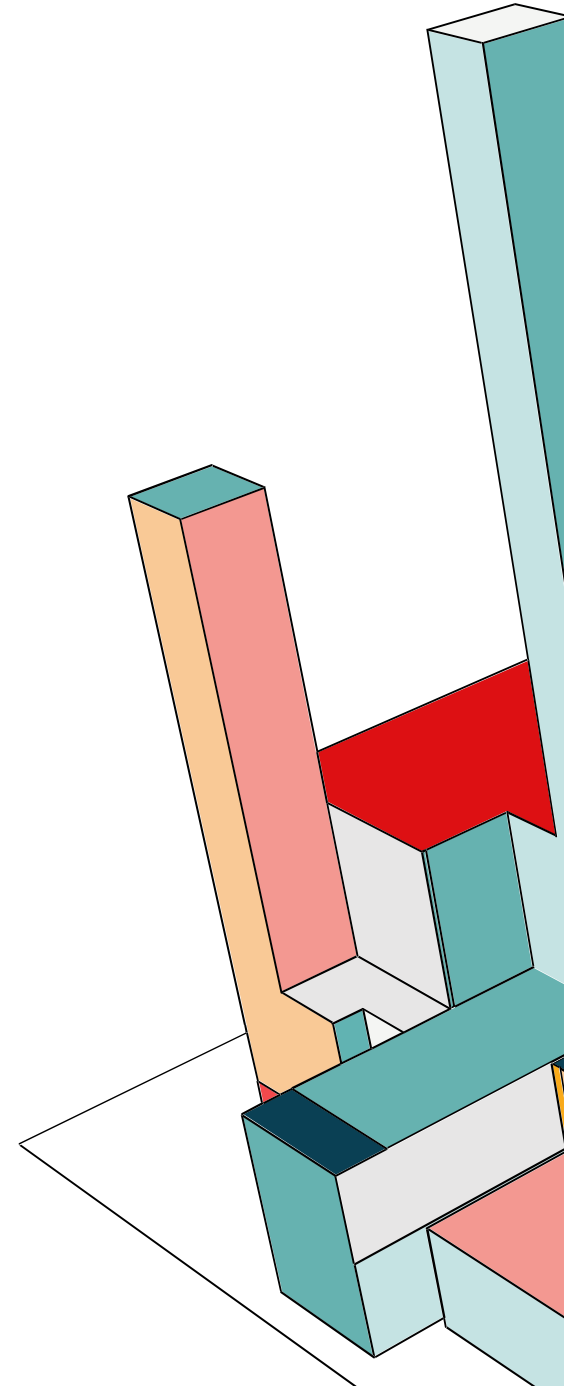
- Integrity Checks
- Remove Null

2. Data Inspection

- Understanding the data
- Removing null values

3. Analysis Preparation

- Pulling data from multiple sheets and merge into one master sheet
- Update new column follow specified rule



ANALYSIS & VISUALIZATION



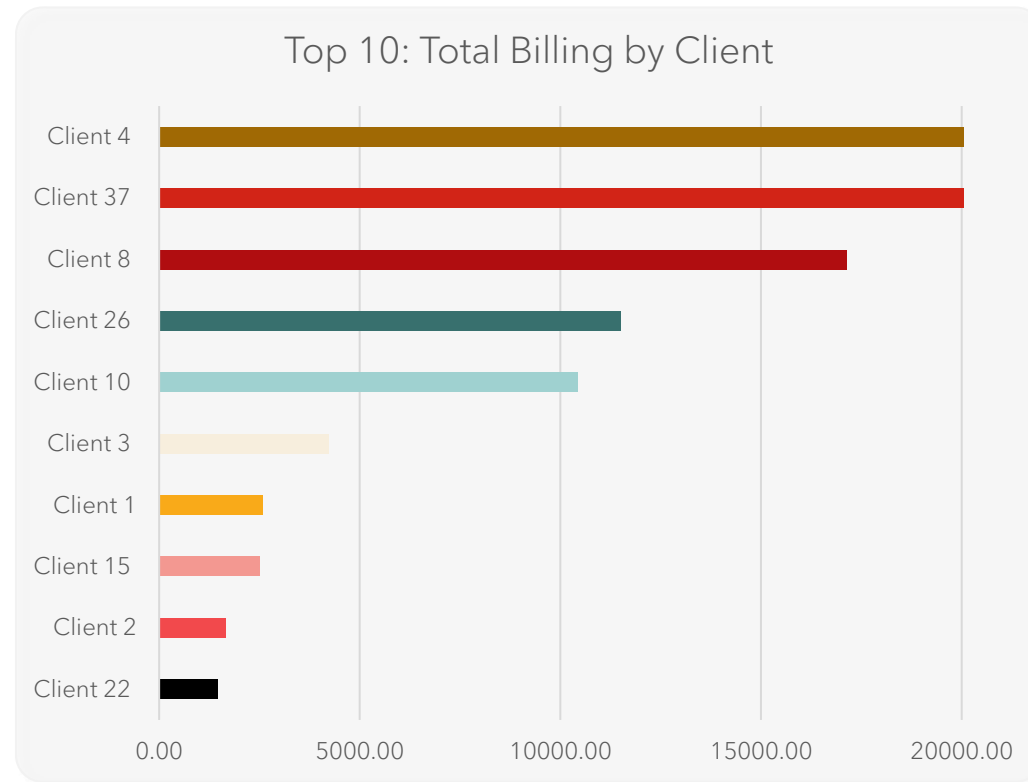
ANALYSIS AND VISUALIZATION

Answering and Visualizations from the data:

- Total Billing per Client
- Gross Margins by Client %
- Gross Margins by Country %
- Gross Margins by Carrier %
- Gross Margins by Number %
- Total Cost per Vendor
- Gross Margin by Country/Carrier/Client

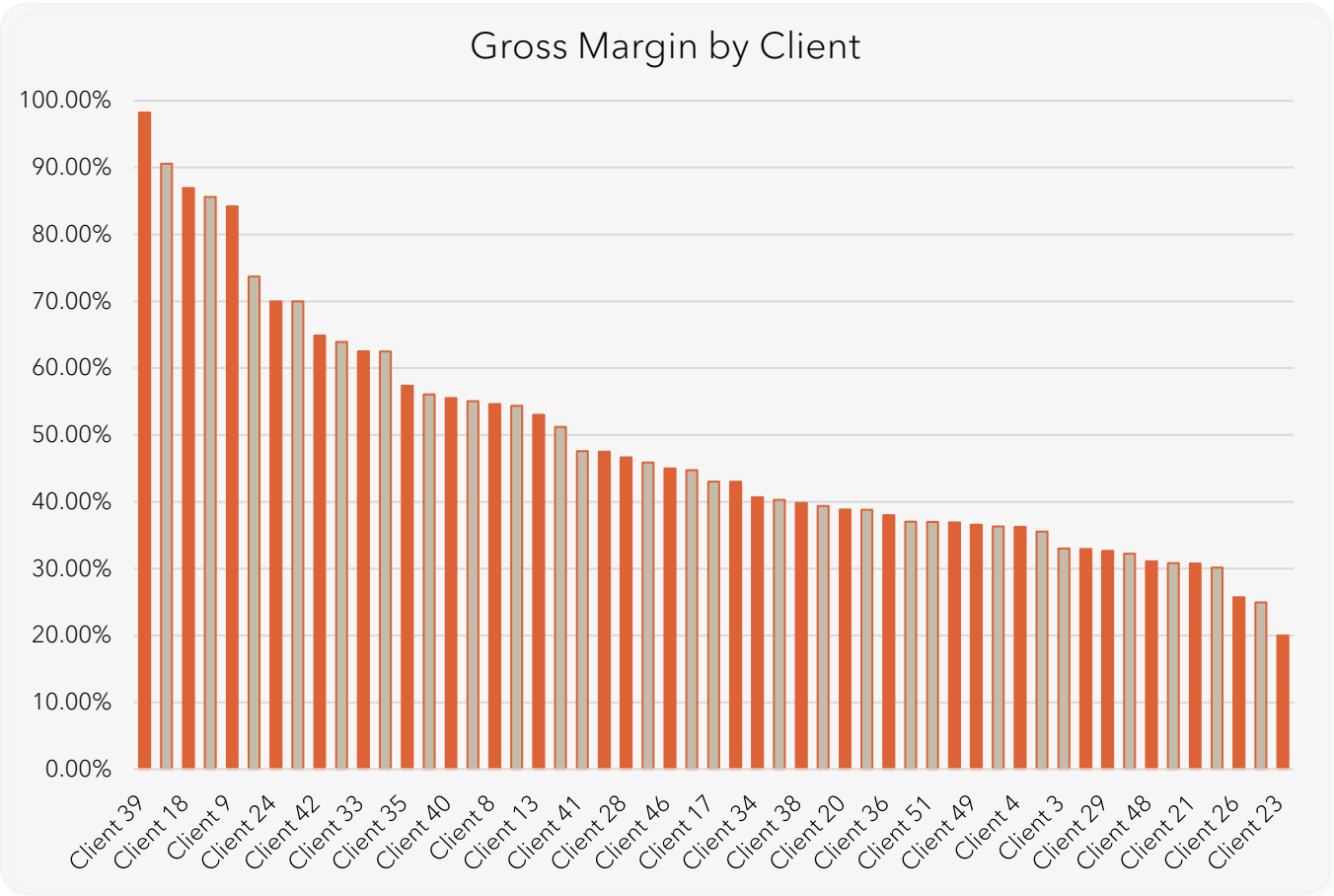
TOTAL BILLING PER CLIENT

Customer	Total Billing per Client
Client 22	1459.01
Client 2	1650.35
Client 15	2497.68
Client 1	2584.75
Client 3	4219.41
Client 10	10418.98
Client 26	11504.19
Client 8	17142.29
Client 37	20384.75
Client 4	20431.43
Grand Total	92292.84



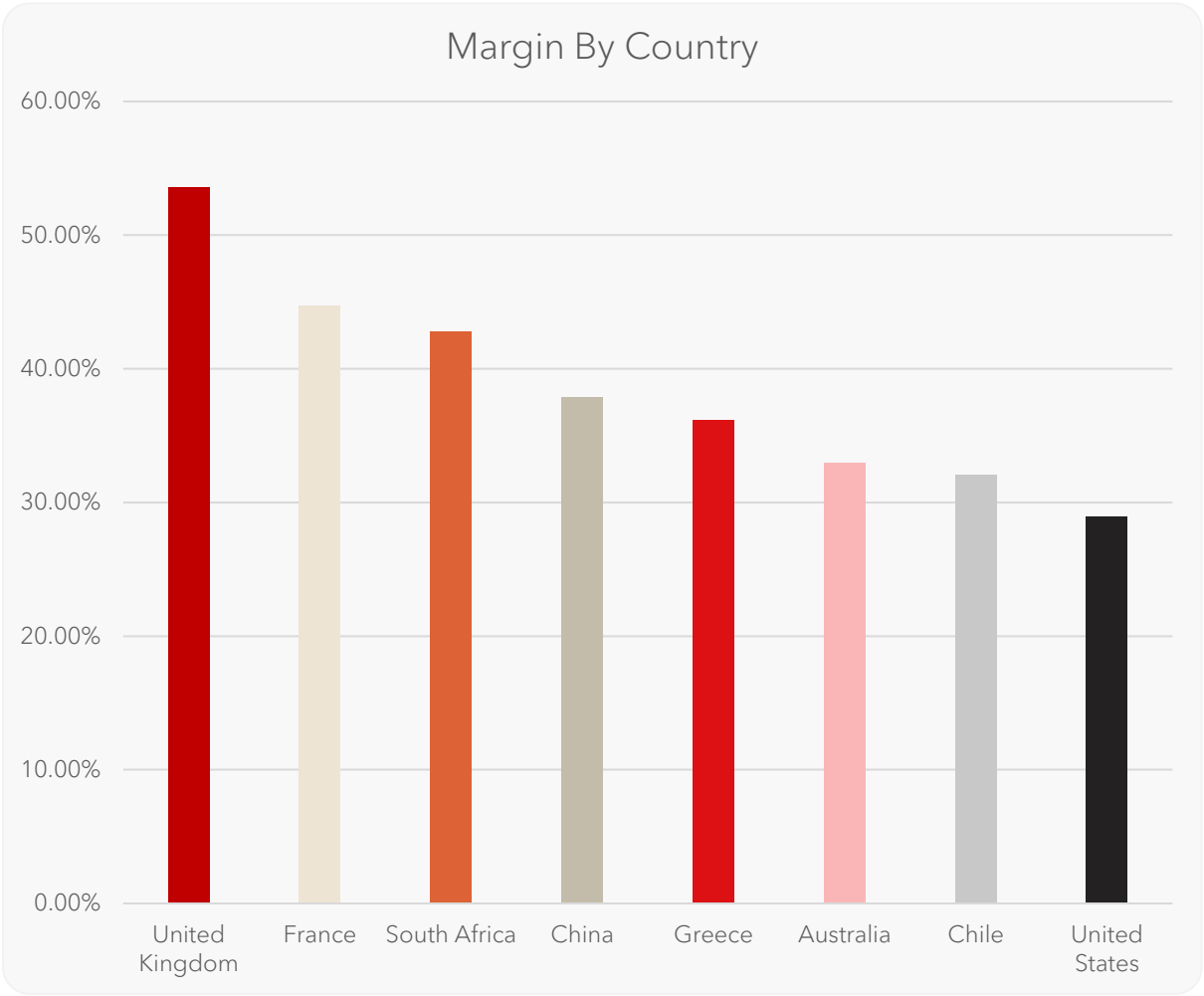
GROSS MARGIN BY CLIENT %

Customer	Gross Margin Percentage
Client 39	98.23%
Client 45	90.59%
Client 18	86.93%
Client 32	85.62%
Client 9	84.20%
Client 7	73.70%
Client 24	70.00%
Client 14	70.00%
Client 42	64.89%
Client 6	63.92%
Client 33	62.52%
Client 43	62.49%
Client 35	57.38%
Client 19	56.07%
Client 40	55.55%
Client 16	55.06%
Client 8	54.61%
Client 44	54.36%
Client 13	53.01%
Client 11	51.20%
Client 41	47.57%
Client 10	47.48%
Client 28	46.66%
Client 30	45.87%
Client 46	45.00%
Client 1	44.74%
Client 17	43.02%
Client 52	43.01%
Client 34	40.68%
Client 47	40.28%
Client 38	39.84%
Client 12	39.38%
Client 20	38.86%
Client 53	38.81%
Client 36	38.00%
Client 5	37.03%
Client 51	36.99%
Client 31	36.92%
Client 49	36.56%
Client 37	36.31%
Client 4	36.24%
Client 25	35.54%
Client 3	33.01%
Client 2	32.93%
Client 29	32.67%
Client 50	32.24%
Client 48	31.10%
Client 15	30.83%
Client 21	30.77%
Client 22	30.16%
Client 26	25.69%
Client 27	24.94%
Client 23	20.00%



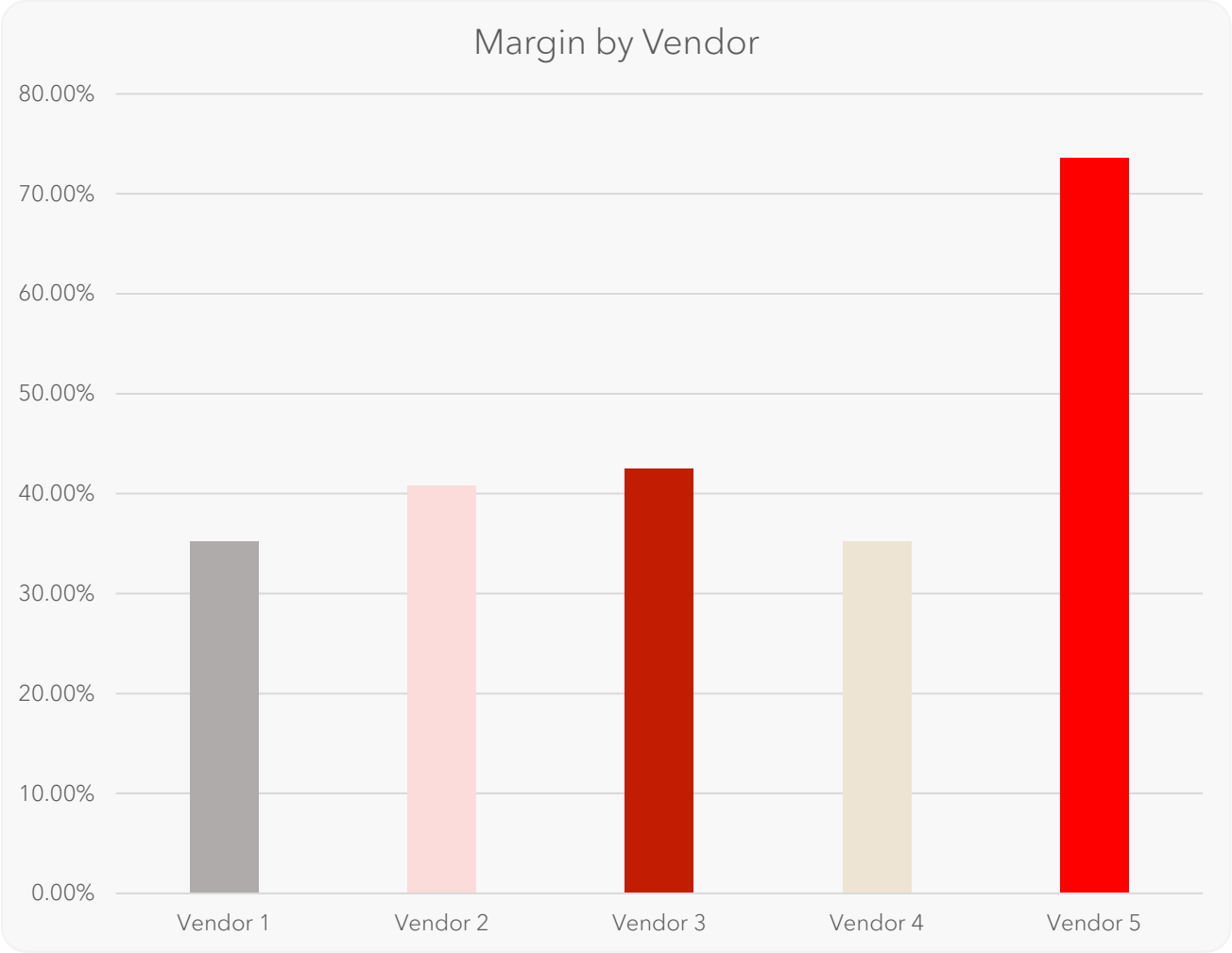
GROSS MARGIN BY COUNTRY %

Country	Gross Margin Percentage
United Kingdom	53.53%
France	44.69%
South Africa	42.80%
China	37.88%
Greece	36.16%
Australia	32.96%
Chile	32.07%
United States	28.95%



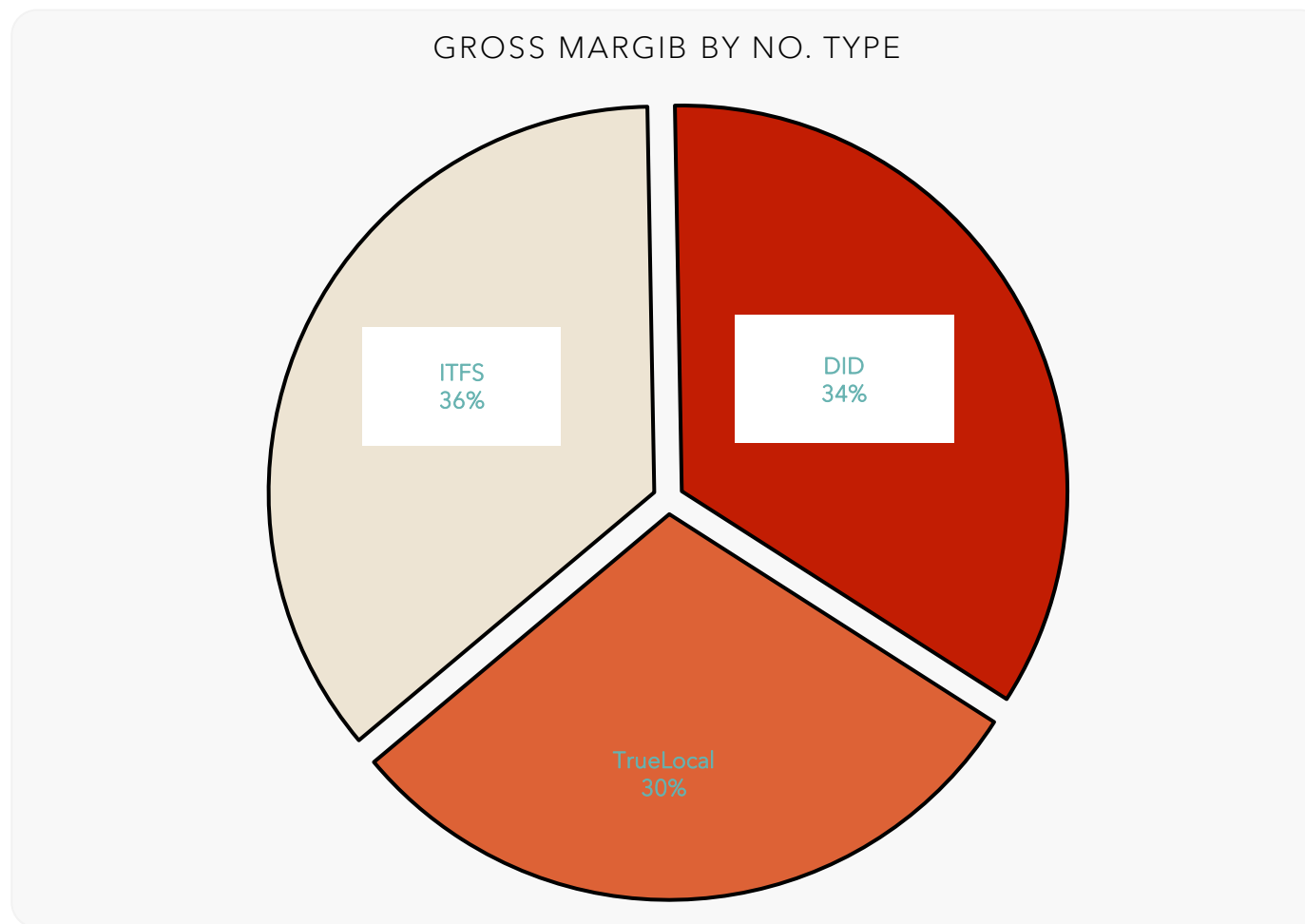
GROSS MARGIN BY VENDOR TYPE %

Vendor Type	Gross Margin Percentage
Vendor 1	35.18%
Vendor 2	40.82%
Vendor 3	42.49%
Vendor 4	35.23%
Vendor 5	73.61%



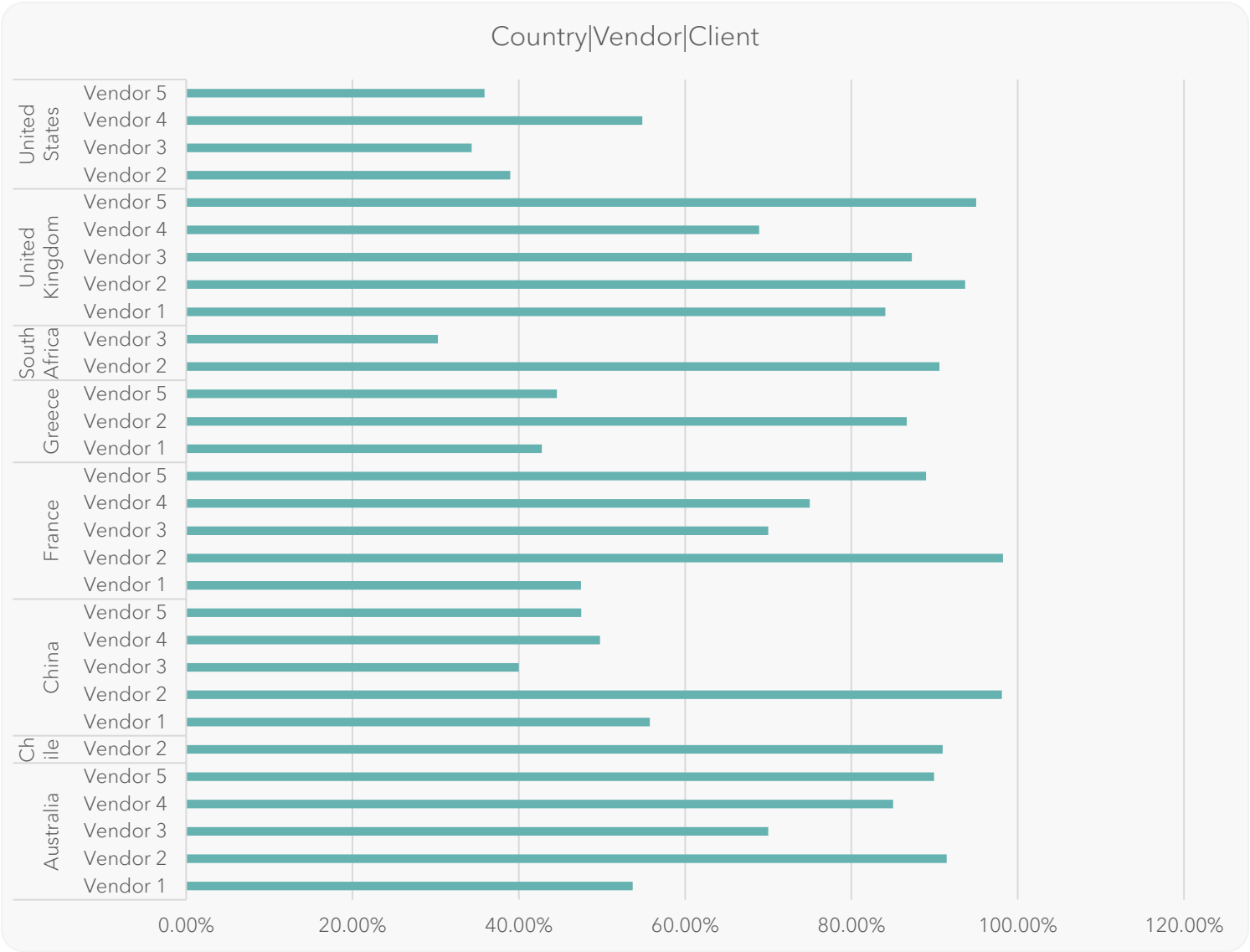
GROSS MARGIN BY NUMBER TYPE %

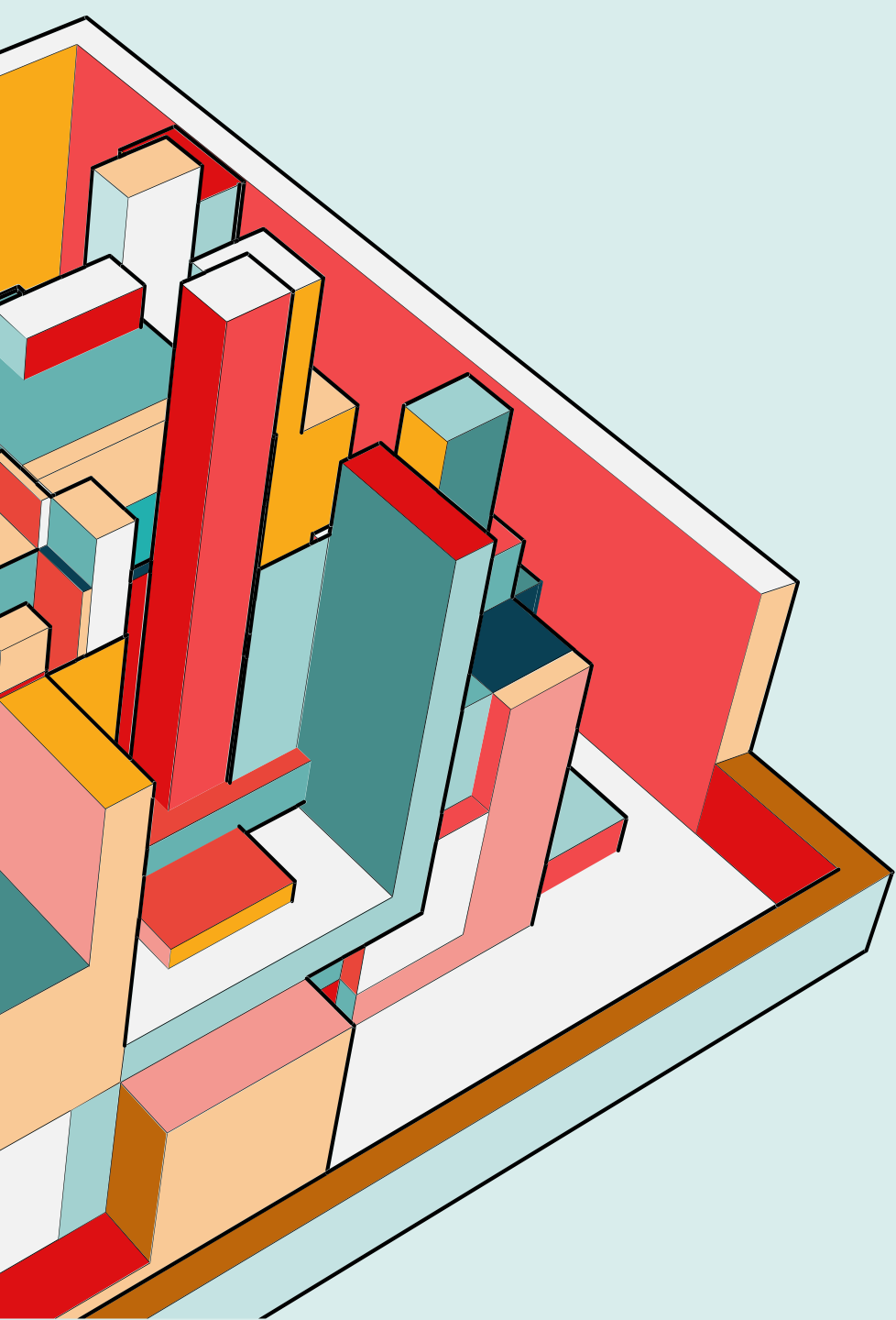
Number Type	Gross Margin Percentage
ITFS	40.63%
DID	38.95%
TrueLocal	33.83%



GROSS MARGIN COUNTRY | VENDOR | CLIENT

Row Labels	Gross Margin Percentage
Australia	77.12%
Vendor 1	53.69%
Vendor 2	91.47%
Vendor 3	70.00%
Vendor 4	85.00%
Vendor 5	89.95%
Chile	90.97%
Vendor 2	90.97%
China	43.82%
Vendor 1	55.75%
Vendor 2	98.08%
Vendor 3	40.00%
Vendor 4	49.75%
Vendor 5	47.50%
France	59.74%
Vendor 1	47.48%
Vendor 2	98.23%
Vendor 3	70.00%
Vendor 4	75.00%
Vendor 5	88.98%
Greece	51.39%
Vendor 1	42.79%
Vendor 2	86.65%
Vendor 5	44.58%
South Africa	38.77%
Vendor 2	90.60%
Vendor 3	30.26%
United Kingdom	73.70%
Vendor 1	84.09%
Vendor 2	93.67%
Vendor 3	87.27%
Vendor 4	68.89%
Vendor 5	95.00%
United States	38.28%
Vendor 2	38.97%
Vendor 3	34.33%
Vendor 4	54.87%
Vendor 5	35.87%



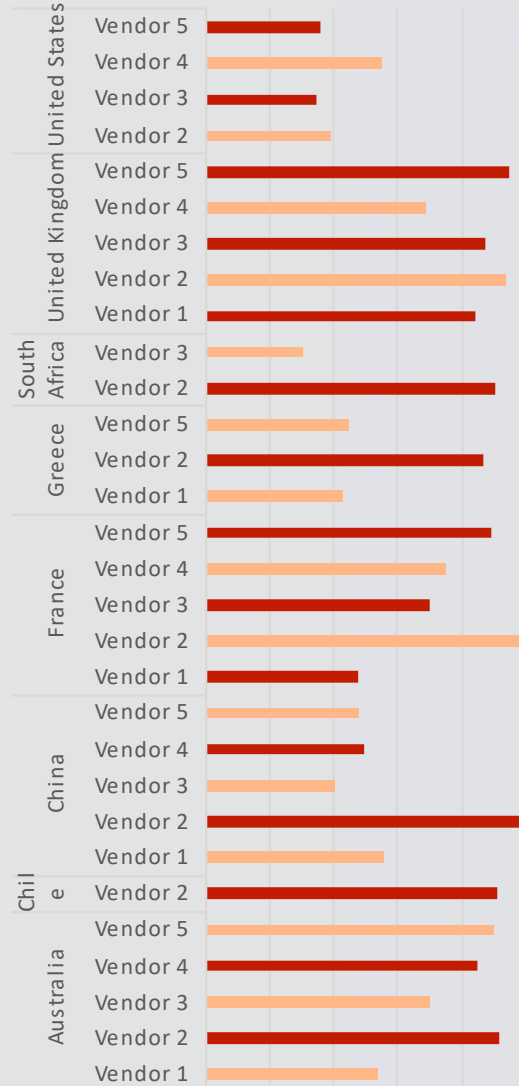


DASHBOARD

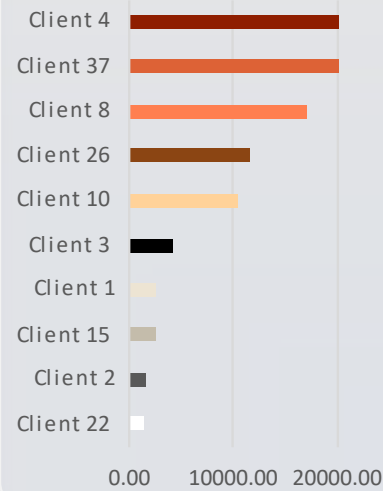


Nasrul Khair

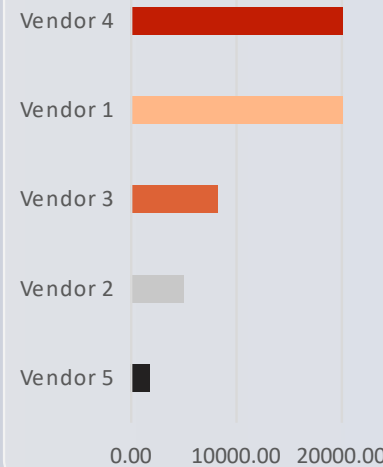
Country|Vendor|Client



Top 10: Revenue



Cost



Total Revenue

\$95,236.83

Total Cost

\$57,503.27

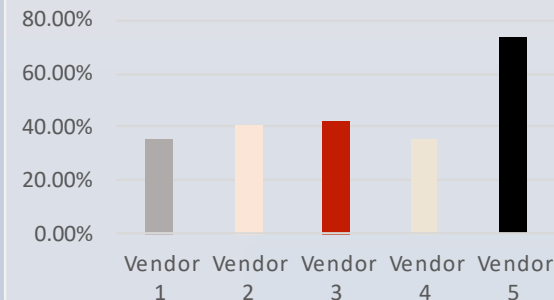
Vendor 2

Australia

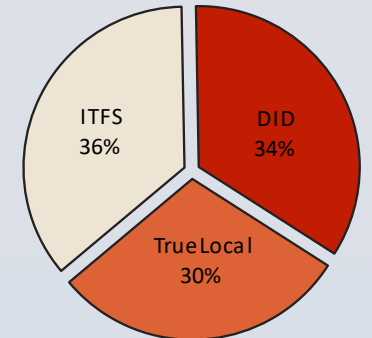
TrueLocal

"Clients 4 and 37 contribute the highest share to the company's revenue, totaling 42.77%. On the vendor side, Vendor 4 represents the largest expense for the company— a stark contrast to Vendor 5, which incurs the least."

Margin by Vendor



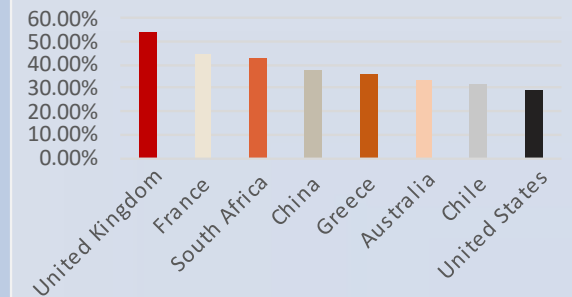
MARGIN
BY
NO. TYPE

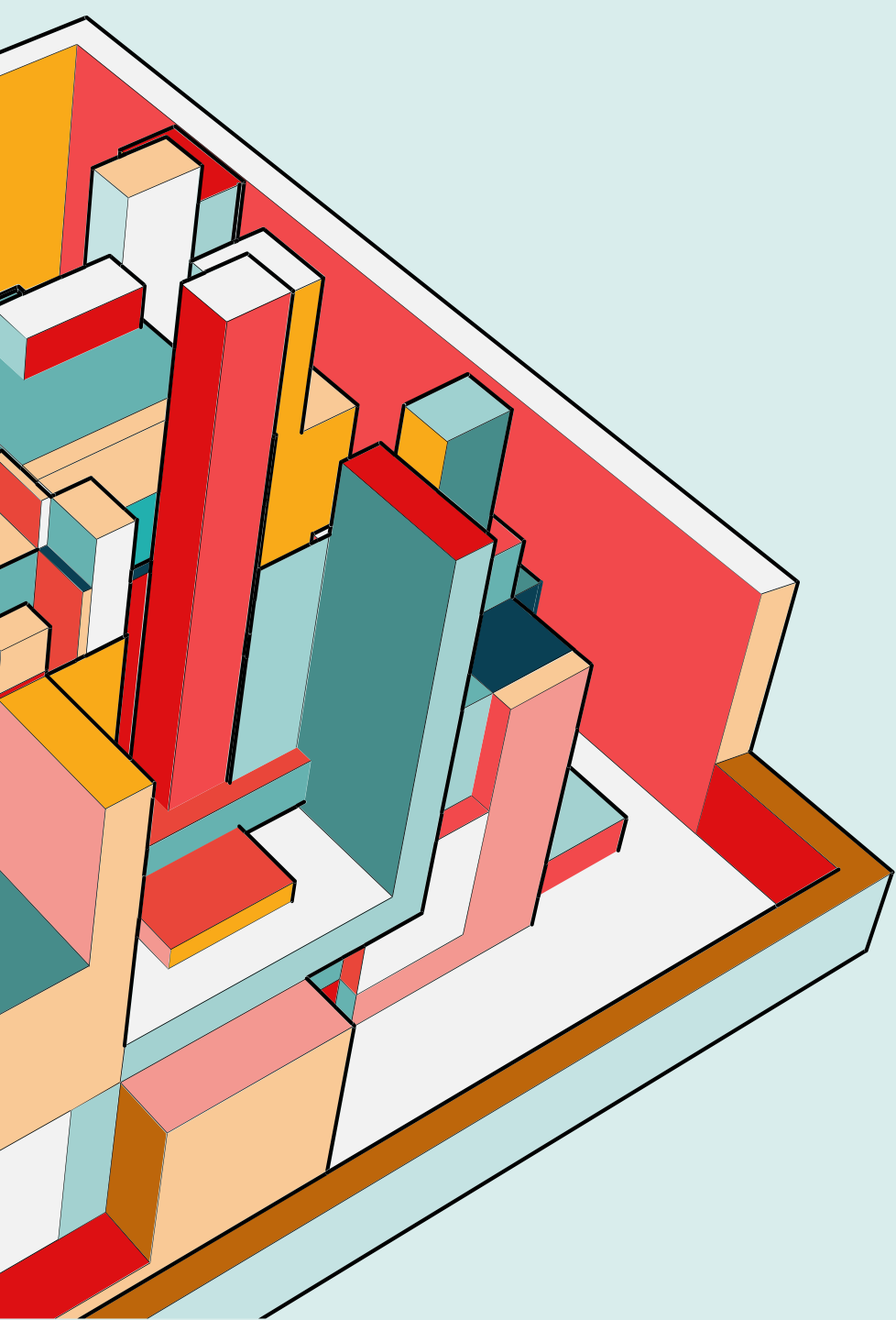


Margin by Client



Margin By Country





AUTOMATION

AUTOMATE: CALCULATION & VISUALIZATION

- VBA :

```
Private Sub Worksheet_Change(ByVal Target As Range)
    Dim pt As PivotTable
    Dim ws As Worksheet

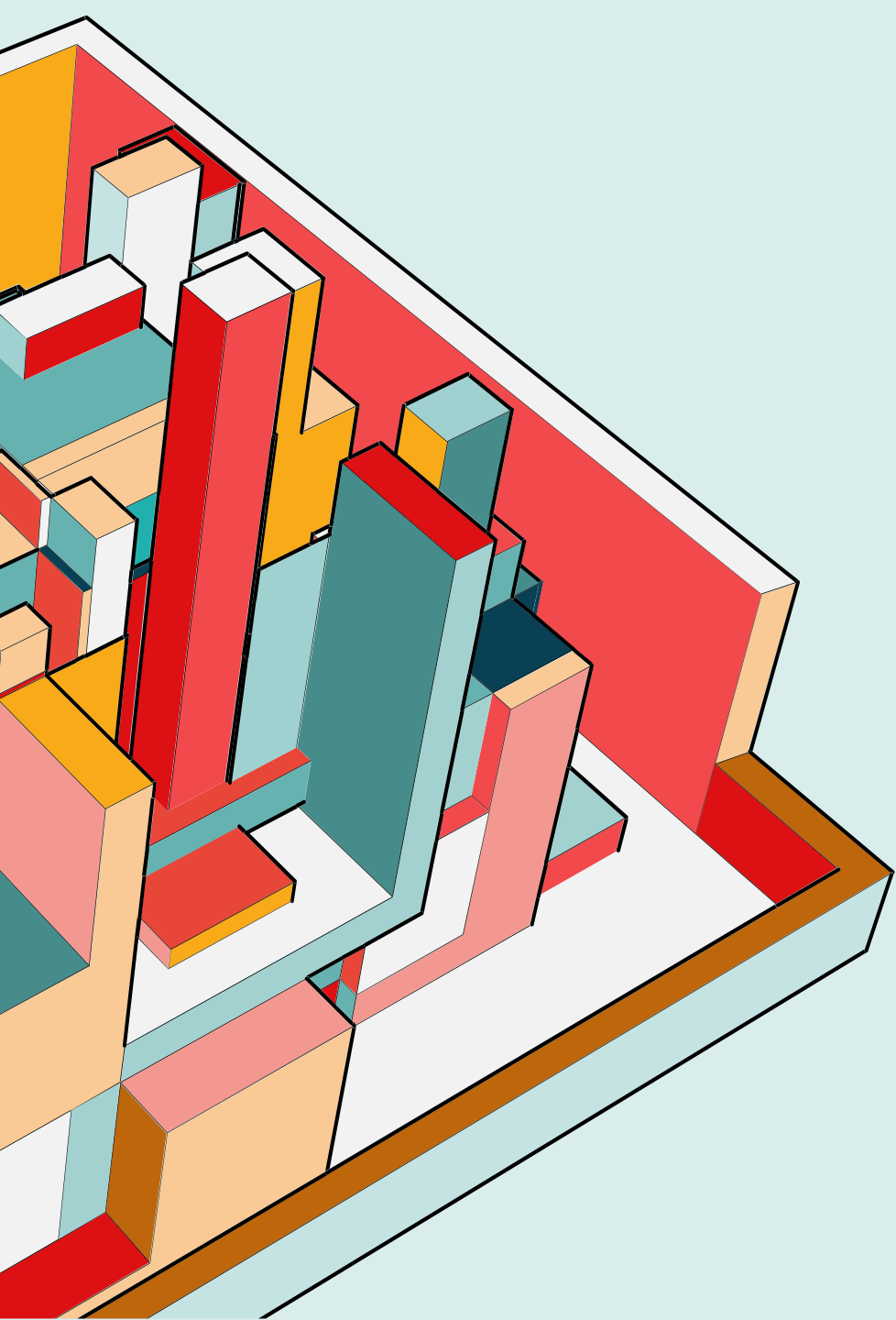
    For Each ws In ThisWorkbook.Worksheets
        For Each pt In ws.PivotTables
            pt.RefreshTable
        Next pt
    Next ws
End Sub
```

- Automate flexibility in testing the rate increment
- Copying into new file 'master_automate' to run as data source for Power BI.
- Table to enter different rate value.

Client Billed (Minutes)	Vendor Type	Vendor Billed (Minutes)	Client Rate	Vendor Rate	Revenue	Cost	Gross Ma
2.00	Vendor 5	2.00	0.0012	0.0008	0.0024	0.0017	0.30
2.00	Vendor 5	1.50	0.0012	0.0008	0.0024	0.0013	0.48
4.00	Vendor 5	4.00	0.0012	0.0008	0.0048	0.0034	0.30
2.00	Vendor 5	1.50	0.0012	0.0008	0.0024	0.0013	0.48
1.00	Vendor 5	1.00	0.0012	0.0008	0.0012	0.0008	0.30
5.00	Vendor 5	5.00	0.0012	0.0008	0.0060	0.0042	0.30
3.00	Vendor 5	2.50	0.0012	0.0008	0.0036	0.0021	0.42
6.00	Vendor 5	6.00	0.0012	0.0008	0.0072	0.0050	0.30
3.00	Vendor 1	2.90	0.1070	0.0749	0.3210	0.2172	0.32
2.00	Vendor 1	1.90	0.1070	0.0749	0.2140	0.1423	0.34
9.00	Vendor 1	8.50	0.1070	0.0749	0.9630	0.6367	0.34
8.00	Vendor 1	7.30	0.1070	0.0749	0.8560	0.5468	0.36
31.00	Vendor 1	30.40	0.1070	0.0749	3.3170	2.2770	0.31
2.00	Vendor 1	1.10	0.1070	0.0749	0.2140	0.0824	0.62
7.00	Vendor 1	6.10	0.1070	0.0749	0.7490	0.4569	0.39
1.00	Vendor 1	0.90	0.1070	0.0749	0.1070	0.0674	0.37

Landline Rate Increase	Mobile Rate Increase
20%	7%





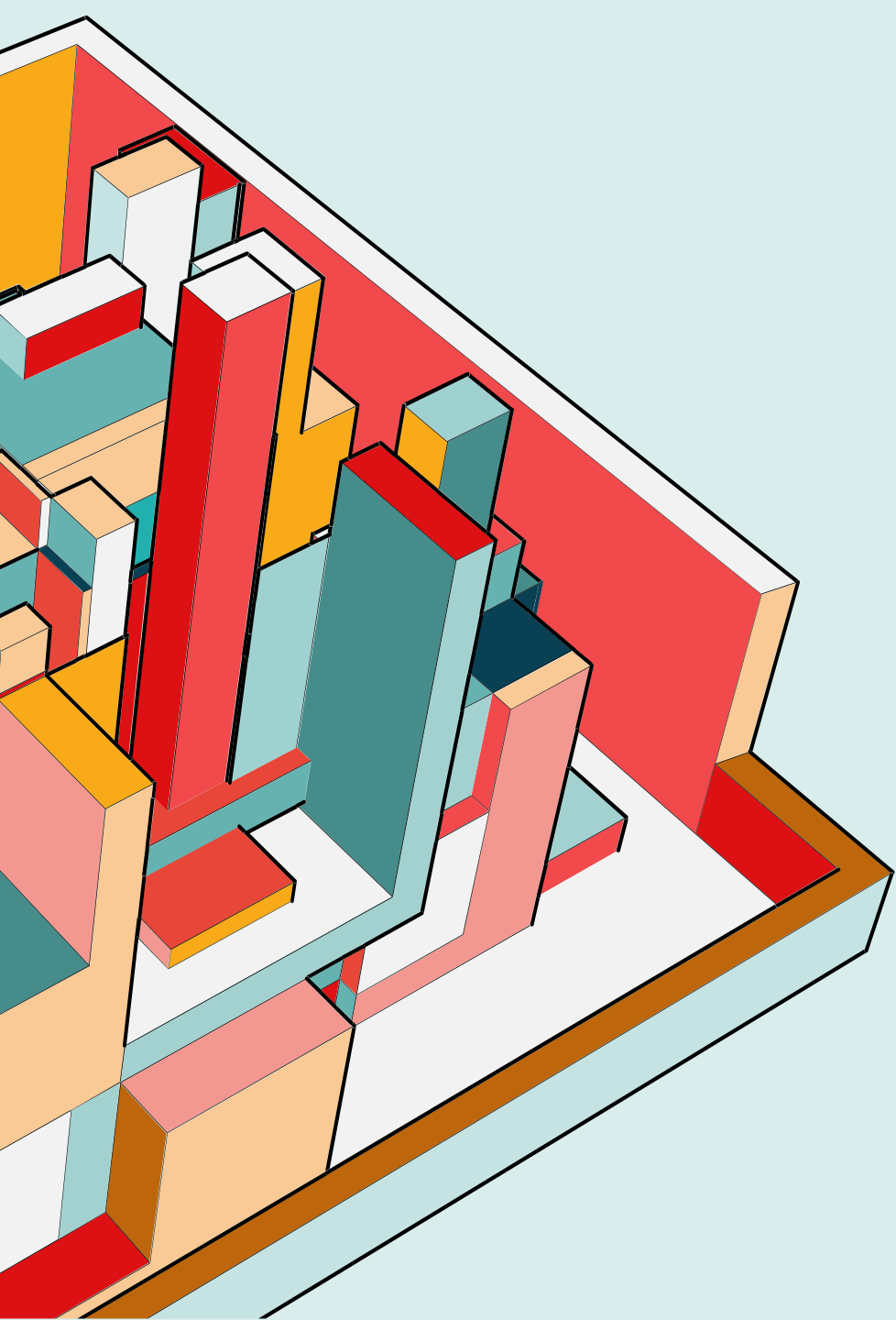
RATE SIMULATION

LANDLINE AND MOBILE RATE INCREMENT

Customer	Total Billing (Client)	Adjusted Billing	Difference
4	\$18,095.18	\$20,431.43	\$2,336.25
37	\$18,065.45	\$20,384.75	\$2,319.30
8	\$15,150.11	\$17,142.29	\$1,992.18
26	\$10,389.24	\$11,504.19	\$1,114.95
10	\$9,210.20	\$10,418.98	\$1,208.78
3	\$3,809.43	\$4,219.41	\$409.98
1	\$2,288.63	\$2,584.75	\$296.12
15	\$2,177.70	\$2,497.68	\$319.98
2	\$1,477.05	\$1,650.35	\$173.30
22	\$1,262.53	\$1,459.01	\$196.48

- Automate flexibility in testing the rate increment
- Splitting into new DataFrame, security purpose and comparison
- Comparing before and after value based on the rate adjusted





BUSINESS QUESTION

BUSINESS QUESTION

Changes if increased the mobile rate by 7% and the landline rate by 20%??

- There is no significance changes towards the gross margin except for Total billing per client and Total Cost per vendor
- Refer table for the difference:

Total Bill per Client

Customer	Total Billing	Total Billing Adjusted	Difference
Client 22	\$ 1,262.53	\$ 1,459.01	\$ 196.48
Client 2	\$ 1,477.05	\$ 1,650.35	\$ 173.30
Client 15	\$ 2,177.70	\$ 2,497.68	\$ 319.98
Client 1	\$ 2,288.63	\$ 2,584.75	\$ 296.12
Client 3	\$ 3,809.43	\$ 4,219.41	\$ 409.98
Client 10	\$ 9,210.20	\$ 10,418.98	\$ 1,208.78
Client 26	\$ 10,389.24	\$ 11,504.19	\$ 1,114.95
Client 8	\$ 15,150.11	\$ 17,142.29	\$ 1,992.18
Client 37	\$ 18,065.45	\$ 20,384.75	\$ 2,319.30
Client 4	\$ 18,095.18	\$ 20,431.43	\$ 2,336.26
Grand Total	\$ 81,925.51	\$ 92,292.84	\$ 10,367.33

Total Cost per Vendor

Vendor	Total Cost	Total Cost Adjusted	Difference
Vendor 5	1639.39	1847.80	208.41
Vendor 2	4483.61	5086.37	602.76
Vendor 3	7186.48	8145.40	958.92
Vendor 1	18487.05	20647.52	2160.46
Vendor 4	19275.99	21776.18	2500.19
Grand Total	51072.52	57503.27	6430.74

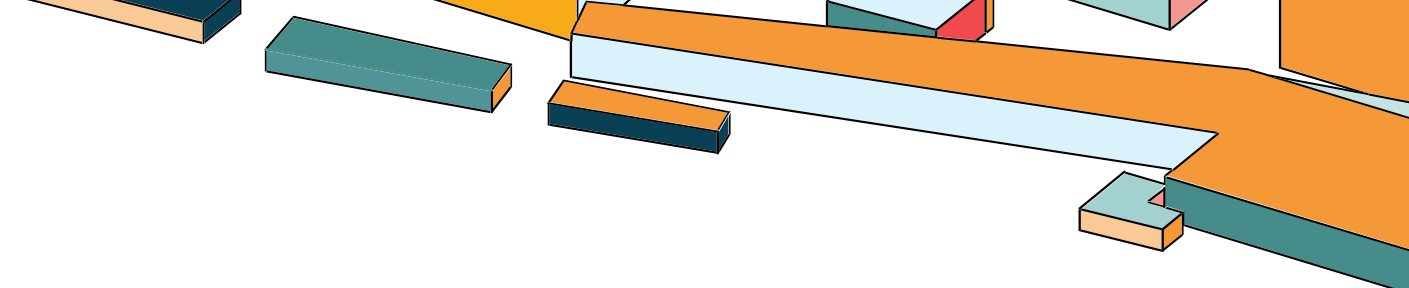


BUSINESS QUESTION

What is this data telling us?

- There is a clear difference between client billing logic and vendor cost structure, which directly affects profitability.
- Certain clients consistently yield lower margins due to:
 - High vendor rates
 - Low client billing rates
 - Short duration calls with inefficient billing increments (e.g., billed per minute but actual call is 10 seconds).
- There is a clear difference between client billing logic and vendor cost structure, which directly affects profitability.





BUSINESS QUESTION:

Changes make to have the overall Gross Margin to be at 45% ?

- To reach a 45% overall gross margin, the company need to reduce the total vendor by approximately **\$4,575.30**
- The estimation of cost reduction per vendor listed as below:

Vendor	Cost Reduction Needed
Vendor 4	2900.05
Vendor 1	2821.73
Vendor 2	314.97
Vendor 3	309.29
Vendor 5	-1770.75

- *Vendor 4 and 1* accounts for ~70% pf the total cost reduction required
- *Vendor 5* aperforming above the required margin with a surplus of approximately **\$1770**
- Vendor 2 and 3 require moderate cost reductions

Conclusion:

- Prioritize renegotiations with Vendor 4 and Vendor 1 to bring their pricing closer to the target vendor cost level.
- Set cost benchmarks internally based on Vendor 5's performance.
- Explore alternative vendors or lower-cost for high-volume destinations.



RECOMMENDATIONS AND RISKS

INSIGHTS

- Significant margin variability across clients and vendors
- Short duration calls impact profitability when billed per minute
- Monthly patterns suggest behavioral or seasonal changes
- Some vendors are costlier for similar destinations

RECOMMENDATIONS AND RISKS

Opportunities

- **Rethink Routing:**
 - Some call routes are costing more than they bring in. It's worth reviewing these and seeing if you can negotiate better deals with vendors or switch to more affordable carriers.
- **Update Client Agreements:**
 - A few clients are using a lot of minutes but paying too little. Their contracts might be outdated and not in line with current costs.
- **Introduce Tiered Pricing:**
 - Consider offering pricing based on usage levels. This way, high-usage clients pay a bit more, helping to balance out the costs and boost profit margins.
- **Adjust for Short Calls:**
 - If a lot of calls are really short, and the billing per second, company could be losing money. Switching to billing in 30-second increments (or similar) could help recover more revenue from those short calls. ("Follow vendor charging method").
- **Rate Adjustment:**
 - Instead of increasing both the landline and mobile rates across the board, the rate adjustments should focus on improving the company's margins—not just passing on more profit to the carrier.
 - For example, raise the rates charged only to the client, not from the vendor. This way, we can avoid a situation where higher vendor costs eat into our revenue and cause a deficit



THANK YOU

Nasrul Khair

017-716 3119

nasrul.bursa@gmail.com

nasrulkhair.github.io

